Intrinsic Defenses Against Backdoor Attacks in High-Order Graph Neural Networks via Semantic and Outlier-Guided Subgraph Policies

Abhijeet Dhali
Department of Computer Science
Brock University
St. Catharines, Canada.
adhali@brocku.ca

Renata Dividino
Department of Computer Science
Brock University
St. Catharines, Canada.
rdividino@brocku.ca

***Abstract*—Graph Neural Networks (GNNs) are increasingly used in sensitive areas such as healthcare and finance, where reliability and security are critical. However, their vulnerability to adversarial backdoor attacks creates serious risks. These attacks insert harmful triggers during training to force the model to produce incorrect, targeted outputs during inference. Although several defense strategies have been proposed, designing GNN systems that are naturally resistant to such attacks remains a significant challenge.**

In this work, we leverage the enhanced expressiveness of higher-order GNNs and a substructure learning approach to propose a new, robust system that includes two built-in defense mechanisms. The first is a substructure extraction method that uses cosine similarity to measure the semantic alignment between connected nodes. Edges between nodes that are semantically different are marked as potential triggers and are removed. As a result, the substructures are formed by more consistent neighborhoods and meaningful relationships, and the model becomes more resistant to backdoor attacks that violate graph homophily. The second mechanism is an outlier detection-based approach that uses clustering to identify dominant and cohesive substructures within the graph. Nodes that differ significantly from these core structures are marked as potential triggers. By isolating and filtering out such anomalies, the model can reduce the success rate of backdoor attacks that introduce unnatural or deceptive graph elements. We evaluate our approach on standard datasets and under various state-of-the-art attack scenarios. Results show a significant reduction in backdoor attack success rates while maintaining high clean accuracy. These findings demonstrate the potential of embedding structure-aware defense mechanisms directly into GNN systems. This project's source code is publicly available at https://github.com/AbhiJeet70/SPROUT_GNN and permanently archived with DOI: 10.5281/zenodo.17095064.

***Index Terms*—Graph Neural Networks (GNNs), Adversarial Backdoor Attacks, Cosine Similarity–based Subgraph Extraction, Clustering-based Outlier Detection, Robustness and Security in GNNs.**

I. Introduction

Graph Neural Networks (GNNs) represent a specialized type of neural network architecture designed to process graph-structured data, where nodes represent entities and edges connections between them. Using their message-passing approach, GNNs repeatedly collect information from neighbor nodes to build rich node representations that capture both node features and local graph patterns [1]. This ability has led to important advances in many fields, from creating models of biological networks for finding new drugs and studying diseases [2], to finding fraud in financial transactions, to providing customized content suggestions on entertainment websites [3].

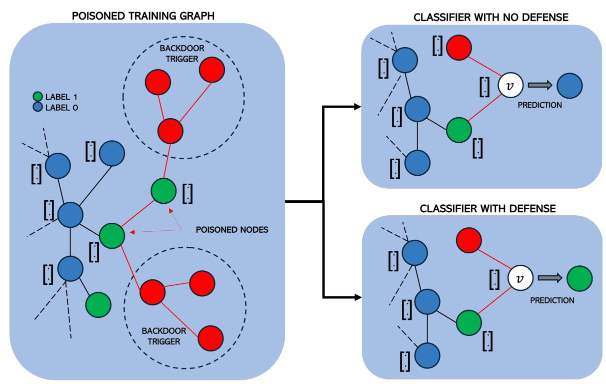

Fig. 1. Backdoor Attack on GNNs

However, GNNs face serious problems with adversarial backdoor attacks, where attackers intentionally change data to deceive models into making wrong or biases predictions. These attacks appear in different ways, such as spammers adding fake accounts to social networks to change community patterns or changing online reviews to affect product ratings. Backdoor attacks on GNNs [4, 5, 6, 7] specifically try to make models incorrectly classify certain nodes, i.e., to infer their labels incorrectly. Adversaries achieve this by adding carefully designed triggers, such as new node connections, that change the local graph structure around target nodes. During training, the model learns to connect these trigger patterns with the attacker's chosen target labels, causing reliable wrong classification when these triggers appear during inference (as shown in Figure 1). The message-passing methods in GNNs make models especially weak to small changes in structure or features that can spread through the network, greatly affecting prediction results.

Current backdoor attacks on GNNs have created different ways to add harmful triggers [4, 5, 6, 7]. A major difficulty in creating effective attacks is finding the best attack budget, which means using the smallest number of harmful triggers while keeping both attack success and the model's normal performance. Because of this, many current attacks add triggers that change some features of the original graph, making them easy to find. For example, [4] adds triggers by connecting nodes that are not similar, which reduces homophily in the graph, making these changes easy to detect by defenses that analyze structure. Other attacks, like those in [5, 6], add triggers that are very different from the original graph structure. These out-of-distribution triggers can usually be found through outlier analysis [8]. But more advanced attacks have appeared that use in-distribution triggers [7] that closely copy real graph patterns, successfully avoiding traditional anomaly detection methods. This development of attack methods shows the strong need for better defense systems to match these complex strategies.

Current defense methods against backdoor attacks in GNNs mainly focus on cleaning the poisoned data before it is used for training by finding anomalies and studying graph structures to identify and remove poisoned training data [6, 8]. While these methods have worked well in defending traditional GNN architectures against such attacks, they are naturally preventive and may not be enough against new attack strategies. Recent research has shown that no single defense method effectively stops all current attack methods [9]. To solve these problems, researchers have started exploring combined defensive strategies within GNN architectures themselves. These include defenses based on deep reinforcement learning [10], edge rewiring methods [11], and approaches that model GNNs as connected dynamical systems [12]. At the same time, higher-order graph neural networks (HOGNNs) have become a promising direction for improving GNN capabilities by capturing polyadic relationships beyond simple pairwise edges. By modeling complex multi-node interactions, HOGNNs effectively solve common problems of standard GNNs, such as over-smoothing and over-squashing, leading to more powerful and scalable architectures. Specifically, subgraph HOGNNs [13, 14, 15] have shown great flexibility in handling complex relational data, thanks to their ability to perform message passing over selected vertex and edge collections, which results in greater robustness against backdoor attacks [16]. Relevant to our approach are recent works that exploit such HOGNN architectures designed with built-in defenses against some existing backdoor attacks [17].

Building on recent advances, we leverage the enhanced expressiveness of HOGNN substructure learning frameworks to propose a novel, robust architecture that integrates two built-in defense mechanisms against backdoor attacks. The first mechanism is a substructure extraction strategy that uses cosine similarity to measure semantic alignment between connected nodes. Edges between semantically dissimilar nodes are pruned, resulting in cleaner and more coherent neighborhoods. This pruning helps the model focus on structurally and semantically meaningful relationships, thereby improving message passing and the overall quality of graph representations. As a result, the model becomes more resilient to backdoor attacks that violate graph homophily or introduce misleading structural patterns. The second mechanism is an outlier detection approach that applies clustering to identify dominant and cohesive substructures within the graph. Nodes that significantly deviate from these core structures are flagged as potential out-of-distribution triggers. By isolating and filtering out such anomalies, the model can detect and mitigate backdoor attacks that inject unnatural or deceptive graph elements.

We conduct extensive experiments on benchmark datasets under a variety of state-of-the-art attack scenarios. Our results show that the proposed method significantly reduces backdoor attack success rates while maintaining high clean accuracy. These findings highlight the promise of embedding structure-aware defense mechanisms directly into GNN architectures, marking an important step toward more robust and reliable models suitable for deployment in security-critical applications.

II. Foundations

We provide the formal definition and terminologies used in this work. Let a graph be represented as $G = (V, E, X, A, Y)$ with set of nodes $v_1, ..., v_N \in V$ and edges $(v_i, v_j) \in E$. $X \in \mathbb{R}^{N \times n}$ is the feature matrix where X_i corresponds to the n-dimensional feature vector of node $v_i \in V$, and an adjacency matrix $A \in \{0, 1\}^{N \times N}$ in which A_{ij} is 1 if there exists an edge between any two nodes $v_i \in V$ and $v_j \in V$ and 0 otherwise. We denote $Y = \{y_1, ..., y_M\}$ as a set of M distinct node labels.

A. Graph Neural Networks

Graph Neural Networks (GNNs) have enabled the training of Deep Learning (DL) models on graph-structured data, aiming to represent graphs as low-dimensional embeddings that can be leveraged for downstream tasks, such as node classification, link prediction, and graph classification. A number of architectures have been proposed [18, 19, 20, 21] following the message-passing scheme as follows: GNNs learn a multiple-step mapping $f : G \rightarrow Z \in \mathbb{R}^{N \times n'}$, where Z is an updated feature matrix of a graph G, to represent nodes and edges G into low-dimensional vectors (i.e., embeddings). Each step typically consists of two phases: 1) aggregate: in the aggregation phase, information about the neighborhood of every node is gathered; 2) update: in the update phase, the features of every node is updated based on its current feature vector and the aggregated neighborhood information. These steps are applied as many times as the number of layers in the GNN. The hidden representation $\mathbf{h}_v^{(k)}$ of a node v at the k-th layer is updated as given: $\mathbf{h}_v^{(k)} = \sigma\left(\mathbf{B}_k \mathbf{h}_v^{(k-1)} + \mathbf{W}_k \sum_{u \in \mathcal{N}(v)} g(\mathbf{h}_u^{(k-1)})\right)$ where the matrices $\mathbf{B}_k$ and $\mathbf{W}_k$ are learnable weight parameters at the k-th layer, $\mathbf{h}_v^{(0)} = x_0$, $\mathcal{N}(v)$ is the set of neighbors of v, g is a differentiable, permutation-invariant neighborhood aggregation function that can take various forms, such as mean, sum, or max pooling. The features of node v and its neighbors are combined by the update function, which is a differentiable function that takes the aggregated embeddings as input and produces a new embedding for each node such as MLPs (multi-layer perceptrons). The output of the final layer Z^k can then be used for downstream tasks. GNNs have achieved state-of-the-art performance in many tasks; however, research has shown [21] that they cannot distinguish certain simple graph structures, limiting their expressivity to that of the 1-WL (Weisfeiler-Lehman) test.

B. Higher Order Graph Neural Networks

Higher Order Graph Neural Networks (HOGNNs) represent a significant advancement in GNN architectures, addressing fundamental limitations of traditional GNNs by incorporating graph substructures into their learning process. Traditional GNNs are constrained by the 1-WL test in their ability to distinguish certain graph structures, particularly struggling to differentiate between graphs that are structurally different but possess similar local neighborhoods. HOGNNs overcome these limitations by explicitly considering larger substructures within the graph, allowing finer-grained structural discrimination.

In particular, subgraph-based models like ESAN [13], SAGNN [15], and SUN [14] generate embeddings from selected subgraphs and integrate them into a unified graph representation. The subgraph selection process, governed by the selection policy $\mathcal{P}$, can be implemented using various methodologies. These include ego networks centered around nodes incorporating all vertices within k-hops, edge deletion approaches that randomly remove edges to create sparser subgraphs, node deletion methods sampling subsets of nodes and their induced subgraphs, motif-based selection focusing on specific structural patterns, and random walk approaches utilizing stochastic traversal to sample connected subgraphs.

The mathematical framework of HOGNNs extends traditional GNN's blueprint with two key additions: subgraph generation and final subgraph aggregation. The process begins with subgraph generation, where $\mathcal{S} = S_i = (V_i, E_i) \mid i = 1, \ldots, k, \quad \text{where } S_i \subseteq G$. This is followed by subgraph embedding through the mapping $f : S_i \rightarrow Z_i \in \mathbb{R}^{N \times n'}$, message passing within each subgraph, and final aggregation expressed as $Z_G = \Phi(Z_1, \ldots, Z_k)$. Within each subgraph, the message passing mechanism incorporates node feature aggregation, edge feature integration when available, and update functions that transform the aggregated information across multiple layers to capture hierarchical structures. The final aggregation function Φ can be implemented through various mechanisms, such as pooling operations, attention mechanisms that weight different subgraph contributions, learnable neural networks combining subgraph representations, and concatenation followed by transformation. This flexibility in aggregation allows HOGNNs to effectively capture and combine information from different structural levels.

C. Backdoor Attacks on GNNs

Backdoor attacks are a specific form of adversarial attack designed to manipulate the training phase of machine learning models by injecting malicious patterns, or *triggers*, into the training data. These attacks are unique because the model performs as expected under regular conditions, but makes targeted misclassifications when presented with the specific trigger patterns during inference. Backdoor attacks in GNNs can target multiple classification tasks: (1) node classification: triggers prompt the GNN to misclassify individual nodes that contain the trigger pattern, and (2) graph classification: entire graphs are misclassified when they contain an embedded trigger subgraph, impacting high-level predictions.

Backdoor attacks for node classification are defined as follows: Let G represent a clean graph. Given a target class y_{target}, the goal of a backdoor attack is to train the model so that it associates a specific trigger pattern T with y_{target}, so that when the pattern is present in an input sample, the model outputs y_{target} regardless of the true label. During the training phase, a subset of nodes $V_{\text{poisoned}} \subset V$ is selected, and a trigger subgraph $T = (V_T, E_T)$ is attached to these nodes. The trigger subgraph is designed so that it has specific properties or patterns (e.g., high cosine similarity to the target node features or unique connectivity patterns). The resulting graph is defined as: $G_{\text{poisoned}} = (V \cup V_T, E \cup E_T)$.

III. Related Work

A. Recent Backdoor Attacks

Subgraph-Based Attack (SBA) [4] introduced SBA, a backdoor attack framework that manipulates GNNs by injecting predefined subgraph triggers into training samples to achieve attacker-specified predictions. SBA can be executed in two primary variants: SBA-Samp and SBA-Gen. SBA-Samp generates fixed subgraph triggers using the Erdős–Rényi (ER) model [22], where edges are formed at a fixed probability and node features are sampled uniformly at random from the training graph. While these universal triggers are easier to implement, they may be more detectable due to their misalignment with natural graph structures. The enhanced variant, SBA-Gen, adaptively generates triggers based on the input graph, creating node features using a Gaussian distribution with mean and variance computed from actual nodes within the graph. This approach produces more realistic and stealthy triggers that blend naturally with the existing data, leading to increased attack success rates. By embedding these carefully crafted subgraphs, SBA creates effective backdoor triggers that prompt the model to predict an attacker-specified target label whenever the trigger appears, while evading statistical anomaly detection during inference.

Graph Trojan Attack (GTA) [5] introduced the GTA, which advances beyond static triggers by implementing an adaptive, sample-specific trigger generation approach. GTA employs a trigger generator, implemented as a two-layer MLP, that creates unique trigger subgraphs for each node or sample in the dataset. This customization of perturbations enhances

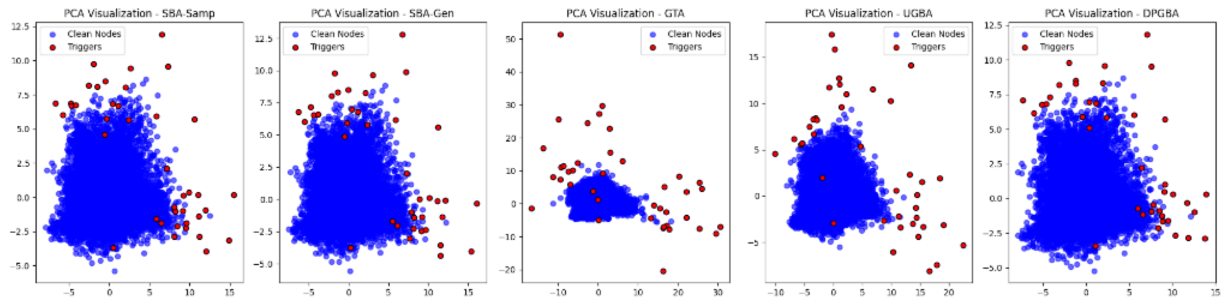

Fig. 2. Preliminary Analysis of Trigger Distribution. Clean nodes are shown in blue, and trigger nodes are shown in red for the attacks SBA, GTA, UGBA, and DPGBA.

the attack's efficacy by allowing the model to learn precise associations between specific triggers and targeted labels. The sample-specific nature of the triggers maximizes both the attack's subtlety and impact. However, this approach, while effective, does not incorporate considerations for attack budget constraints or maintain similarity metrics between the original and perturbed graphs.

Unnoticeable Graph Backdoor Attack (UGBA) [6] proposed UGBA, which advances backdoor attack by focusing on the *unnoticeability* of trigger patterns. UGBA employs a sophisticated approach that maximizes the cosine similarity between the feature vectors of triggers and target nodes, expressed mathematically as $max(cos(\mathbf{h}_T, \mathbf{h}_v))$, where $(\mathbf{h}_T)$ and $(\mathbf{h}_v)$ represent the feature representations of trigger and target nodes, respectively. The attack utilizes a trigger generator while strategically selecting target nodes that will maximize the attack's impact within a constrained budget. By ensuring triggers are both structurally and feature-wise similar to natural graph patterns, UGBA achieves seamless integration of malicious elements into the graph structure, making detection significantly more challenging while maintaining attack effectiveness.

Distribution Preserving Graph Backdoor Attack (DPGBA) [7] developed the DPGBA, which introduces a sophisticated approach to generating in-distribution (ID) triggers that effectively mimic legitimate data patterns. DPGBA employs a combination of adversarial learning and an out-of-distribution (OOD) detector implemented as an autoencoder. The reconstruction loss calculated between the encoder and decoder guides the generation of triggers that align closely with the data's natural distribution. This approach is particularly effective at evading outlier-based defenses, which typically identify backdoor triggers by detecting nodes that don't belong to the cluster of most cohesive nodes relevant to the model's classification task. By maintaining the original data distribution while embedding backdoor triggers, DPGBA significantly enhances the attack's stealthiness while preserving its effectiveness.

Effective & Unnoticeable Multi-Category Backdoor Attack (EUMC) [23] creates a Multi-Category Subgraph Triggers Pool (MC-STP) by sampling hundreds of subgraphs using Breadth-First Search around central nodes and scores each one using Attachment Probability Shift (APS) calculated by a clean surrogate GCN to tackle the problem of multi-category backdoor attacks. For each poison node, they select the trigger whose subgraph features best match the node via cosine similarity and attach it by connecting the single most similar node plus all subgraph nodes that surpass a similarity threshold.

B. Backdoor Attack Defense Mechanisms

Defense mechanisms against GNN backdoor attacks typically fall into two main categories: anomaly detection approaches that identify suspicious outliers in the data and structural analysis methods that detect violations of graph homophily, where connected nodes tend to share similar properties. These complementary approaches leverage different aspects of graph data to identify potential backdoor triggers. For our analysis, we selected one representative defense mechanism from each category to examine in detail.

SimGuard [24] introduced SimGuard, a two-stage defense that (1) uses DBSCAN clustering on raw features in conjunction with a global anomaly score based on Canberra distance and autoencoder-based reconstruction screening to flag anomalous nodes and (2) trains a contrastive-learning detector on filtered clean and trigger candidates to learn embeddings that distinguish genuine nodes from triggers. A lightweight MLP classifier is then used to remove triggers precisely, achieving near-perfect detection recall/precision and complete recovery of poisoned nodes while maintaining clean accuracy.

DMGNN: Detecting and Mitigating Backdoor Attacks in Graph Neural Networks [25] introduce DMGNN, which uncovers both out-of-distribution and stealthy in-distribution triggers through counterfactual explanation: diffusing the backdoored graph to a noise level and then denoising to generate minimal perturbations that flip target-class predictions, thereby pinpointing trigger subgraphs. A subsequent reverse sampling pruning step iteratively isolates and removes the most trigger-representative edges, lowering attack success rates and keeping accuracy reasonable.

Outlier Detection [8] proposed *Simultaneous Dominant Sets Clustering (DOMINANT)*, a well-known unsupervised outlier detection approach based on autoencoders for graph data, which uses reconstruction error from both graph structure

TABLE I
PRELIMINARY ANALYSIS OF THE EFFECTIVENESS OF STATE-OF-THE-ART DEFENSE MECHANISMS FOR TRADITIONAL GNN ARCHITECTURES (HERE **GCN MODELS**) ON CORA, PUBMED, AND CITESEER. EACH CELL SHOWS ASR (↓) (TOP) AND ACC (↑) (BOTTOM).

	ASR	**ACC**	**ASR**	**ACC**	**ASR**	**ACC**	**ASR**	**ACC**	**ASR**	**ACC**
Dataset: Cora	SBA-Samp		SBA-Gen		GTA		UGBA		DPGBA	
Defence-None	90.00	**88.54**	100.00	**87.43**	90.00	**87.43**	100.00	**87.80**	100.00	**87.62**
Defence-Dominant Set	90.00	79.30	100.00	78.00	90.00	78.74	80.00	78.37	90.00	79.11
Defence-Prune	90.00	83.92	90.00	81.70	100.00	79.30	80.00	79.85	80.00	79.11
Defence-Prune+LD	**0.00**	46.40	**0.00**	44.92	**0.00**	43.99	**0.00**	44.55	**0.00**	44.18
Dataset: PubMed	SBA-Samp		SBA-Gen		GTA		UGBA		DPGBA	
Defence-None	90.00	**83.39**	92.50	**85.39**	92.50	**86.03**	95.00	**86.46**	95.00	**86.63**
Defence-Dominant Set	82.50	74.87	85.00	76.62	47.50	76.77	87.50	77.76	42.50	77.38
Defence-Prune	92.50	81.44	95.00	82.04	42.50	82.15	92.50	82.65	60.00	82.73
Defence-Prune+LD	**0.00**	42.43	**0.00**	42.78	**0.00**	42.96	**0.00**	43.19	**0.00**	43.52
Dataset: CiteSeer	SBA-Samp		SBA-Gen		GTA		UGBA		DPGBA	
Defence-None	90.00	**75.19**	90.00	**73.38**	90.00	**73.38**	90.00	**73.53**	90.00	**72.78**
Defence-Dominant Set	80.00	67.82	80.00	65.86	90.00	66.17	60.00	64.81	90.00	65.26
Defence-Prune	70.00	72.33	70.00	69.62	80.00	68.72	70.00	67.37	80.00	67.22
Defence-Prune+LD	**0.00**	37.44	**0.00**	36.84	**0.00**	37.14	**0.00**	36.69	**0.00**	37.74

and node features to detect outliers. The basic idea is that the autoencoder will perform better at reconstructing samples that are similar to most of the training data (expected to be normal data) and perform worse at reconstructing outliers. This method further leverages clustering techniques to identify and isolate dominant set-cohesive substructures in the graph, thereby enabling the detection and removal of injected out-of-distribution triggers, as these triggers significantly deviate from the graph's dominant structure or patterns.

Structural Analysis-Based Defense [6] proposed *Prune*, which addresses the detection of backdoor attacks focusing on the homophily property, that is, the tendency for connected nodes to exhibit similar characteristics. This defense removes edges that link dissimilar nodes, effectively disrupting connections that may serve as backdoor triggers. The method uses cosine similarity to evaluate the connection strength between nodes, removing edges between nodes that fail to meet a similarity threshold. An enhanced version, *Prune+LD* (Label Discard), discards both the edges and labels of dissimilar nodes, further strengthening its defense capabilities against in-distribution attacks. By aligning with expected graph patterns, Prune improves model robustness against a wide range of backdoor attacks, particularly those that violate the homophily assumption.

IV. PRELIMINARY ANALYSIS

Following the approach in [7], we examine whether triggers created by current graph backdoor attack methods can be identified by conducting experiments on the Pubmed [26] dataset. We first use existing backdoor attack to insert backdoors into the graph in a semi-supervised setting. Next, we apply Principal Component Analysis (PCA) to reduce the number of dimensions in node features for both normal nodes and trigger nodes, then display them in a 2-dimensional space as illustrated in Figure 2. In this visualization, blue dots represent normal nodes and red dots represent the created triggers. The figure clearly shows that the triggers produced by GTA [5] and UGBA [6] are notably different and distant from the normal data distribution, demonstrating that triggers created by many current algorithms can be identified as outliers (i.e., out-of-distribution triggers), while triggers produced by SBA [4] and DPGBA [7] follows closely the data distribution (i.e., in-distribution triggers). Defense mechanisms must be able to detect both in-distribution and out-of-distribution triggers, which differ significantly in nature.

To assess the impact of attacks and the effectiveness of defense mechanisms, we selected two advanced methods designed to counter such attacks on GNNs: DOMINANT [8] and Prune [6] on GTA, UGBA, SBA (two variants), and DPGBA attacks. We use Attack Success Rate (ASR), defined as the proportion of nodes with the trigger attached that are misclassified into the target class, and Clean Accuracy (ACC), which measures the model's performance on clean test nodes. See Section VI for more details on the evaluation metrics.

Preliminary experimental results on Cora [26], CiteSeer [26], and PubMed [26] datasets are presented in Table I and Table II. The target models are based on GCN [19] and SAGNN [15] architectures. GCN serves as a representative of traditional GNN architectures, while SAGNN exemplifies high-order GNNs. Although, we evaluate a single architecture of each type, recent studies have shown that no existing defense method can universally mitigate all known backdoor attacks on traditional GNNs [9]. In contrast, high-order GNNs (HOGNNs), which perform message passing over selected vertex and edge substructures, demonstrate greater robustness against such attacks [16]. In our experiments, the trigger size is limited to three nodes. We discard the top 3% of samples with the highest reconstruction loss. The tables also report the accuracy of the backdoored GNNs on the clean test set to illustrate the impact of defense methods on predictive performance. As a baseline, we include the accuracy of each model on a clean (non-attacked) graph.

From the tables I and II, we observe that both GTA and UGBA exhibit high attack success rates when no defense is applied. However, the outlier detection approach successfully removes all their triggers, reducing the attack success rate to zero. For SBA and DPGBA, which generate in-distribution

TABLE II

PRELIMINARY ANALYSIS OF THE EFFECTIVENESS OF STATE-OF-THE-ART DEFENSE MECHANISMS FOR HIGH-ORDER GNN ARCHITECTURES (HERE **SAGNN MODELS**) ON CORA, PUBMED, AND CITESEER. EACH CELL SHOWS ASR (↓) (TOP) AND ACC (↑) (BOTTOM).

	ASR	**ACC**	**ASR**	**ACC**	**ASR**	**ACC**	**ASR**	**ACC**	**ASR**	**ACC**
Dataset: Cora	SBA-Samp		SBA-Gen		GTA		UGBA		DPGBA	
Defence-None	**0.00**	**86.88**	**0.00**	**87.25**	20.00	85.40	20.00	85.21	**0.00**	**87.99**
Defence-Dominant Set	**0.00**	85.75	**0.00**	85.20	20.00	86.91	20.00	**87.40**	**0.00**	86.27
Defence-Prune	**0.00**	75.79	**0.00**	75.42	20.00	74.31	20.00	74.31	**0.00**	74.49
Defence-Prune+LD	**0.00**	87.06	10.00	87.11	**10.00**	**88.42**	**0.00**	85.03	**0.00**	87.41
Dataset: PubMed	SBA-Samp		SBA-Gen		GTA		UGBA		DPGBA	
Defence-None	0.83	**87.29**	**0.00**	**87.41**	3.33	**87.44**	5.00	**87.34**	0.83	**87.51**
Defence-Dominant Set	0.50	86.80	0.20	86.95	2.50	86.90	4.00	86.75	0.60	87.00
Defence-Prune	1.00	86.50	0.10	86.70	3.00	86.55	4.50	86.60	0.70	86.80
Defence-Prune+LD	**0.40**	86.90	0.05	87.00	**2.00**	86.95	**3.50**	86.85	**0.50**	87.10
Dataset: CiteSeer	SBA-Samp		SBA-Gen		GTA		UGBA		DPGBA	
Defence-None	**0.00**	71.13	3.33	71.43	3.33	**72.18**	3.33	72.78	**0.00**	71.43
Defence-Dominant Set	**0.00**	73.17	3.33	**73.66**	3.33	70.93	3.33	70.06	3.33	**72.88**
Defence-Prune	**0.00**	68.12	**0.00**	68.12	**0.00**	68.57	**0.00**	69.02	**0.00**	68.27
Defence-Prune+LD	**0.00**	**73.53**	3.33	72.56	3.33	70.88	**0.00**	**73.60**	6.67	72.04

triggers, the attack success rate remains low even though their triggers are not flagged as outliers. The pruning strategy is more effective at identifying these triggers, as some violate the homophily property. Therefore, Prune and DOMINANT detect different types of attacks and can be used to complement each other's strengths. Based on these observations, we propose a robust GNN that leverages the high expressiveness of HOGNNs to preserve accuracy, while integrating Prune and DOMINANT defense mechanisms as built-in components (and rather than relying on preprocessing steps before training).

V. METHODOLOGY

In this section, we introduce SPROUT-GNN, a structure-aware HOGNN architecture with built-in defenses against backdoor attacks. The model's robustness is enhanced through two novel subgraph extraction policies that capture subtle variations in node relationships. Our approach is inspired by the defense mechanisms in Prune [6] and DOMINANT [8]. However, rather than applying these defenses as preprocessing steps on potentially poisoned data, we embed them directly into the model architecture. Drawing on HOGNNs and architectures designed with built-in defenses against certain backdoor attacks [17], we develop an inherently robust system that outperform current existing systems.

A. SPROUT-GNN Extraction Policies

We propose two structure-aware subgraph extraction policies that exploit semantic and structural consistency across neighborhoods to isolate meaningful substructures and suppress adversarial patterns.

The first policy leverages cosine similarity to quantify semantic alignment between neighboring nodes. Let $\mathbf{x}_i \in \mathbb{R}^d$ denote the feature vector of node i, and define its normalized representation as $\tilde{\mathbf{x}}_i = \mathbf{x}_i / \|\mathbf{x}_i\|_2$. For each edge (i, j), the cosine similarity is computed as:

$$\cos(i, j) = \tilde{\mathbf{x}}_i^\top \tilde{\mathbf{x}}_j.$$

Then, for each node i, we compute the average cosine similarity across its neighborhood $\mathcal{N}_i$:

$$\bar{s}_i = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \cos(i, j).$$

Nodes with average similarity scores above a chosen threshold (e.g., the global median) are retained:

$$\theta = \text{median}\left(\{\bar{s}_i\}_{i=1}^n\right), \quad \mathcal{V}^{\text{cos}} = \{i \mid \bar{s}_i \geq \theta\}.$$

Only edges between retained nodes are preserved, yielding a filtered subgraph $G^{\text{c}} = (\mathcal{V}^{\text{c}}, \mathcal{E}^{\text{c}})$. This strategy filters out semantically inconsistent nodes and promotes message passing through coherent and meaningful local structures.

The second policy identifies dominant substructures by assessing each node's proximity to the centroid of its semantic cluster. Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ denote the matrix of node features. The feature space is partitioned into k clusters, where k is typically set to the number of ground-truth classes. Each node i is assigned to a cluster with centroid $c(i)$, and its Euclidean distance is computed as:

$$d_i = \|\mathbf{x}_i - c(i)\|_2.$$

Nodes with distances below a chosen threshold (e.g., the global median) are retained:

$$\phi = \text{median}\left(\{d_i\}_{i=1}^n\right), \quad \mathcal{V}^{\text{dom}} = \{i \mid d_i \leq \phi\}.$$

Edges between retained nodes define the dominant substructure subgraph $G^{\text{d}} = (\mathcal{V}^{\text{d}}, \mathcal{E}^{\text{d}})$. This method preserves the internal structure of dominant clusters while eliminating outlier nodes, which are more likely to correspond to poisoned inputs or adversarial triggers.

B. SPROUT-GNN Architecture

As illustrated in Figure 3, our system incorporates a widely adopted strategy in HOGNN frameworks: ego subgraph. For a given node i, its k-hop ego subgraph is defined as:

$$G_i^{\text{e}} = (\mathcal{V}_i^{\text{e}}, \mathcal{E}_i^{\text{e}}), \quad \mathcal{V}_i^{\text{e}} = \{j \in \mathcal{V} \mid \text{dist}(i, j) \leq k\}.$$

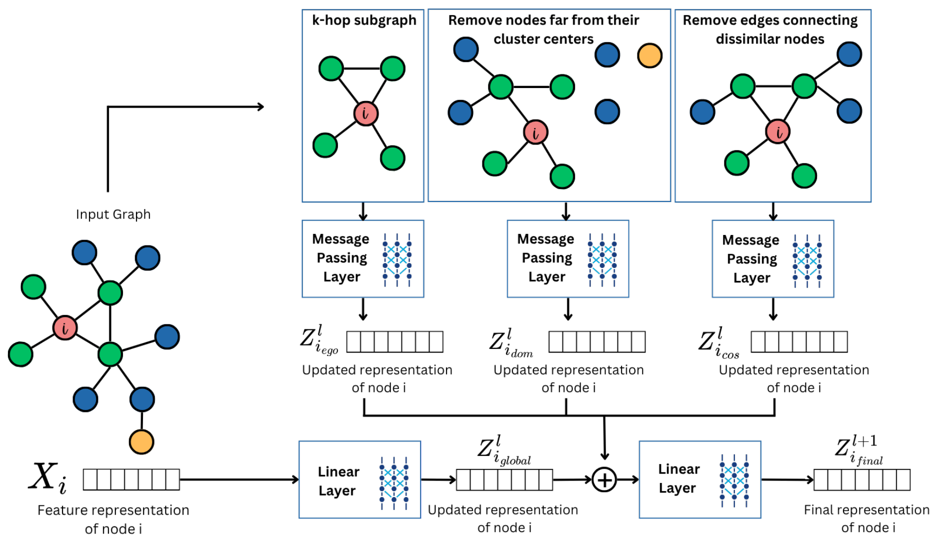

Fig. 3. Substructure-Aware GNN with Cosine Similiarity and Outlier detection (SPROUT-GNN). The system processes the input graph through three parallel subgraph construction strategies: (1) ego-network extraction, which preserves immediate neighborhood information; (2) edge-similatiry-based pruning which removes connections between nodes with dissimilar node representations; and (3) dominant-substructure-alignment pruning which removes triggers significantly deviate from the graph's dominant structure or patterns. These subgraphs undergo separate message-passing operations, while the original graph's node features are transformed through global encoding. The final node representations are generated by combining these three substructure-specific features with the global encoding, creating the final node representation used for prediction.

Here, $\mathcal{E}_i^{\mathrm{e}}$ includes all edges between nodes in $\mathcal{V}_i^{\mathrm{e}}$, and typically $k = 2$. This ego subgraph captures the local topology around node i, supporting higher-order message passing within a structurally localized context.

Each subgraph G (cosine-based, dominant-based, or ego-based) is processed by an independent message-passing layer, yielding:

$$\mathbf{Z}^{\mathrm{e}} = \mathrm{MP}(G^{\mathrm{e}}, A), \quad \mathbf{Z}^{\mathrm{c}} = \mathrm{MP}(G^{\mathrm{c}}, A), \quad \mathbf{Z}^{\mathrm{d}} = \mathrm{MP}(G^{\mathrm{d}}, A).$$

A global encoder applies a linear transformation to the original features:

$$\mathbf{Z}^{\mathrm{global}} = \mathbf{W}_g \mathbf{X} + \mathbf{b}_g,$$

where $\mathbf{W}_g \in \mathbb{R}^{h \times d}$ and $\mathbf{b}_g \in \mathbb{R}^h$ are learnable parameters.

The final node representation is the concatenation of these encoded features along with the global encoding:

$$\mathbf{Z}^{\mathrm{concat}} = \left[\, \mathbf{Z}^{\mathrm{e}} \,\|\, \mathbf{Z}^{\mathrm{c}} \,\|\, \mathbf{Z}^{\mathrm{d}} \,\|\, \mathbf{Z}^{\mathrm{global}} \,\right].$$

The concatenated feature vector is then passed through a fully connected layer to produce the class logits:

$$\hat{\mathbf{Y}} = \mathbf{W}_f \mathbf{Z}^{\mathrm{concat}} + \mathbf{b}_f \in \mathbb{R}^C,$$

where C is the number of classes, and the final class probabilities are computed via softmax.

VI. Evaluation

A. Datasets

We utilize three benchmark citation network datasets: Cora [26], CiteSeer [26], and PubMed [26], each varying in the number of nodes, edges, and feature dimensions, providing diverse structural and attribute characteristics for evaluation.

Dataset	Nodes	Classes	Edges	Dimension
Cora	2,708	7	5,429	1,433
CiteSeer	3,327	6	4,732	3,703
PubMed	19,717	3	44,338	500

B. Baseline Models and Backdoor Attacks

We evaluate our model's resilience against: traditional GNNs (GCN [19], GAT [20], GraphSAGE [18]); subgraph-based HOGNNs (ESAN [13], SUN [14], SAGNN [15], and SAGNN+CS [17]); and under adversarial scenarios (SBA-Samp [4], SBA-Gen [4], GTA [5], UGBA [6], DPGBA [7]).

C. Metrics

Attack Success Rate (ASR): The effectiveness of a backdoor attack is measured by the proportion of nodes with the trigger T attached that are misclassified into the target class y_{target} [4]. For a set of test nodes V_{test}, the ASR is calculated as follow where $\mathbb{I}$ is an indicator function that returns 1 if the model's prediction for v is y_{target} and 0 otherwise:

$$\mathrm{ASR} = \frac{1}{|V_{\text{test}}|} \sum_{v \in V_{\text{test}}} \mathbb{I}(f(G_{\text{poisoned}}, x_v) = y_{\text{target}})$$

Clean Accuracy (ACC): Backdoor attacks should not degrade the model's performance on non-triggered data [4]. ACC measures the model's performance on a set of clean test

TABLE III
ANALYSIS OF THE EFFECTIVENESS OF STATE-OF-THE-ART DEFENSE MECHANISMS FOR VARIOUS GNN ARCHITECTURES ON CORA, CITESEER, AND PUBMED. EACH CELL SHOWS ASR (↓) (TOP) AND ACC (↑) (BOTTOM).

		ASR	**ACC**	**ASR**	**ACC**	**ASR**	**ACC**	**ASR**	**ACC**	**ASR**	**ACC**
Dataset: Cora		SBA-Samp		SBA-Gen		GTA		UGBA		DPGBA	
GCN	Traditional	96.67	87.00	96.67	86.44	70.00	86.14	96.67	85.95	96.67	85.89
GraphSage	Traditional	96.67	87.49	96.67	87.43	86.67	87.37	96.67	87.18	96.67	87.25
GAT	Traditional	93.33	85.46	90.00	85.46	93.33	84.84	90.00	85.03	90.00	84.97
SUN	Higher-Order	**0.00**	86.32	**0.00**	86.14	**10.00**	86.88	**10.00**	87.31	**0.00**	87.31
ESAN (ego)	Higher-Order	**0.00**	44.79	**0.00**	45.16	20.00	44.42	**10.00**	44.79	**0.00**	44.55
ESAN (edge del)	Higher-Order	3.33	85.58	**0.00**	85.71	20.00	85.95	**10.00**	85.64	**0.00**	85.52
ESAN (node del)	Higher-Order	3.33	85.83	**0.00**	85.34	20.00	85.64	**10.00**	85.89	**0.00**	85.95
SAGNN	Higher-Order	**0.00**	85.89	**0.00**	85.83	**10.00**	85.58	**10.00**	86.14	**0.00**	85.58
SAGNN+CS	Higher-Order	**0.00**	87.19	**0.00**	**87.49**	**10.00**	86.69	**10.00**	**87.37**	**0.00**	86.82
Ours SPROUT-GNN	Higher-Order	**0.00**	**87.80**	**0.00**	87.62	20.00	**88.35**	**10.00**	86.88	**0.00**	**87.43**
Dataset: CiteSeer		SBA-Samp		SBA-Gen		GTA		UGBA		DPGBA	
GCN	Traditional	95.56	72.88	96.67	72.43	90.00	72.98	96.67	**73.33**	96.67	73.38
GraphSage	Traditional	93.33	**76.84**	93.33	**76.69**	92.22	76.04	94.44	75.74	93.33	75.74
GAT	Traditional	95.56	72.63	95.56	72.83	93.33	72.78	94.44	72.63	94.44	72.73
SUN	Higher-Order	**0.00**	75.49	1.11	75.64	2.22	75.24	3.33	75.84	2.22	76.04
ESAN (ego)	Higher-Order	1.11	29.62	**0.00**	29.87	**0.00**	29.82	3.33	29.47	3.33	29.83
ESAN (edge del)	Higher-Order	**0.00**	72.73	**0.00**	72.68	**0.00**	72.78	3.33	72.97	**1.11**	72.63
ESAN (node del)	Higher-Order	2.22	73.08	1.11	72.70	**0.00**	72.53	3.33	72.78	**1.11**	72.63
SAGNN	Higher-Order	2.22	75.89	1.11	76.44	3.33	75.94	3.33	**76.49**	2.22	**76.34**
SAGNN+CS	Higher-Order	1.11	76.09	1.11	76.24	3.33	**76.19**	3.33	76.04	3.33	75.94
Ours SPROUT-GNN	Higher-Order	**0.00**	72.18	**0.00**	71.73	6.67	72.33	**0.00**	71.43	**3.33**	72.78
Dataset: PubMed		SBA-Samp		SBA-Gen		GTA		UGBA		DPGBA	
GCN	Traditional	95.00	86.35	95.00	86.89	37.50	86.36	95.00	87.15	94.17	87.33
GraphSage	Traditional	95.00	87.77	94.17	88.16	25.83	88.08	95.00	88.44	95.83	88.48
GAT	Traditional	96.67	87.20	95.83	86.95	87.50	86.14	95.00	86.83	95.83	86.93
SUN	Higher-Order	0.83	**88.27**	**0.00**	**88.49**	5.83	88.44	4.17	**88.66**	**0.00**	**88.70**
ESAN (ego)	Higher-Order	8.33	35.40	3.33	35.24	15.83	35.70	11.67	35.41	5.83	35.99
ESAN (edge del)	Higher-Order	1.67	87.44	**0.00**	87.05	**2.50**	87.51	**2.50**	87.07	**0.00**	87.58
ESAN (node del)	Higher-Order	2.50	87.04	1.67	86.86	**2.50**	87.77	**2.50**	87.92	**0.00**	87.37
SAGNN	Higher-Order	0.83	87.29	**0.00**	87.41	3.33	87.44	5.00	87.34	0.83	87.51
SAGNN+CS	Higher-Order	**0.00**	87.45	0.83	87.44	4.17	87.58	3.33	87.49	**0.00**	87.32
Ours SPROUT-GNN	Higher-Order	**0.00**	87.12	**0.00**	87.41	3.33	**88.58**	3.33	87.92	**0.00**	87.49

nodes V_{clean}, and it is defined as follow where y_v is the true label for node v:

$$\text{ACC} = \frac{1}{|V_{\text{clean}}|} \sum_{v \in V_{\text{clean}}} \mathbb{I}(f(G, x_v) = y_v)$$

D. Results

Table III presents a comprehensive comparison of our SPROUT-GNN against various GNN under diverse backdoor attack scenarios. For the Cora dataset, traditional GNN architectures showed high vulnerability to attacks, with ASR ranging from 70–97% across attack types. HOGNNs demonstrated significantly better robustness, particularly SUN and SAGNN+CS, which achieved 0% ASR against multiple attacks (SBA-Samp, SBA-Gen, and DPGBA). ESAN variants showed mixed results, with some achieving low ASR (0–20%) but at the cost of much lower accuracy. SAGNN+CS maintained a strong accuracy baseline (86.82–87.49%) with robust defense, achieving 0% ASR against SBA-Samp, SBA-Gen, and DPGBA, and 10% ASR against GTA and UGBA. Our proposed SPROUT-GNN demonstrated complete immunity (0% ASR) to SBA-Samp, SBA-Gen, and DPGBA, minor vulnerability (10–20% ASR) to GTA and UGBA, and maintained the highest accuracy (86.88–88.35%). Similar trends appeared in the CiteSeer dataset, where traditional GNNs remained highly vulnerable (ASR 90–97%), and HOGNNs like SAGNN and SAGNN+CS achieved low ASR (0–3.33%) with good accuracy (75.74–76.49%). SPROUT-GNN showed minimal vulnerability, with mostly 0% ASR except for GTA (6.67%) and DPGBA (3.33%), and accuracy between 71.43–72.78%. For the PubMed dataset, traditional GNNs were again vulnerable (ASR mostly above 90%). Higher-order architectures such as SUN and SAGNN+CS demonstrated excellent robustness with very low ASR (0–5.83%). Notably, SUN achieved near-perfect defense with ASR close to 0% against most attacks and the highest accuracy (88.27–88.70%). SPROUT-GNN similarly performed well, with 0% ASR for SBA-Samp, SBA-Gen, and DPGBA, slight vulnerability to GTA and UGBA (3.33%), and strong accuracy around 87.12–87.92%. Overall, SPROUT-GNN consistently offers strong defense across all datasets, balancing ASR reduction and clean accuracy effectively, despite differences in dataset scale.

VII. ABLATION STUDIES

To evaluate the effectiveness of different components in our SPROUT-GNN framework, we conduct comprehensive ablation studies examining various subgraph feature extraction policies on the Cora and CiteSeer datasets. Table IV presents the ASR and ACC results for each policy across multiple experimental runs.

TABLE IV
ABLATION FOR DIFFERENT SUBGRAPH FEATURE EXTRACTION POLICIES ON CORA AND CITESEER DATASETS.

Policy	**ASR**	**ACC**	**ASR**	**ACC**	**ASR**	**ACC**	**ASR**	**ACC**	**ASR**	**ACC**
Dataset: Cora										
Ego k-hop Neighborhood	**0.0**	86.32	**0.0**	85.95	30.0	85.21	**0.0**	86.88	**0.0**	85.58
Dominant Strategy	**0.0**	76.52	**0.0**	77.45	30.0	74.31	20.0	78.19	**0.0**	76.89
Cosine Strategy	**0.0**	74.68	**0.0**	74.12	50.0	72.83	10.0	75.05	**0.0**	74.31
Ego+Dominant Strategy	**0.0**	**87.99**	**0.0**	**88.17**	**20.0**	87.62	**0.0**	86.69	**0.0**	87.62
Ego+Cosine Strategy	**0.0**	86.69	**0.0**	87.25	**20.0**	86.88	20.0	87.06	**0.0**	87.43
Cosine+Dominant Strategy	**0.0**	77.63	**0.0**	77.63	30.0	77.26	10.0	76.89	**0.0**	79.30
Ours SPROUT-GNN	**0.0**	87.80	**0.0**	**88.17**	**20.0**	**88.35**	**0.0**	**87.06**	**0.0**	**87.62**
Dataset: CiteSeer										
Ego k-hop Neighborhood	**0.0**	71.28	**0.0**	70.68	10.0	70.98	**0.0**	72.93	**3.33**	71.43
Dominant Strategy	**0.0**	68.42	**0.0**	68.12	**6.67**	68.42	**0.0**	69.17	**3.33**	69.17
Cosine Strategy	3.33	66.77	**0.0**	67.22	13.33	66.32	**0.0**	67.07	**3.33**	68.42
Ego+Dominant Strategy	**0.0**	71.28	**0.0**	71.43	13.33	71.58	11.11	72.18	**3.33**	71.88
Ego+ Cosine Strategy	**0.0**	71.13	**0.0**	70.68	**6.67**	71.13	22.22	70.23	**3.33**	71.43
Cosine+Dominant Strategy	**0.0**	67.82	3.33	67.52	10.0	68.12	**0.0**	69.47	**3.33**	69.02
Ours SPROUT-GNN	**0.0**	**72.18**	**0.0**	**71.73**	**6.67**	**72.33**	**0.0**	**72.93**	**3.33**	**72.78**

The individual strategy performance analysis reveals distinct characteristics for each approach. The ego k-hop neighborhood strategy demonstrates strong defensive capabilities with consistently low ASR values, achieving 0.0% in most experimental configurations across both datasets. On Cora, this strategy maintains accuracy ranging from 85.21% to 86.88%, while on CiteSeer, the accuracy spans 70.68% to 72.93%. However, some vulnerability is observed in specific scenarios, with ASR reaching 30.0% on Cora under certain conditions.

The dominant strategy alone shows moderate performance with generally low ASR values but reduced accuracy compared to the ego neighborhood approach. On Cora, accuracy ranges from 74.31% to 78.19%, while on CiteSeer, it achieves 68.12% to 69.17%. This strategy exhibits some vulnerability with ASR values up to 30.0% on Cora and 6.67% on CiteSeer, indicating limitations when used in isolation.

The cosine strategy demonstrates the highest vulnerability among individual approaches, with ASR reaching 50.0% on Cora and 13.33% on CiteSeer. The accuracy performance is also the lowest among individual strategies, ranging from 72.83% to 75.05% on Cora and 66.32% to 68.42% on CiteSeer. These results suggest that relying solely on cosine similarity for subgraph feature extraction may compromise both robustness and accuracy. The combination of different strategies generally yields superior performance compared to individual approaches. The ego+dominant strategy combination shows significant improvement, achieving excellent accuracy on Cora (86.69% to 88.17%) with relatively low ASR values (0.0% to 20.0%). On CiteSeer, this combination maintains competitive accuracy (71.28% to 72.18%) with ASR values ranging from 0.0% to 13.33%. The ego+cosine strategy provides balanced performance with accuracy ranging from 86.69% to 87.43% on Cora and 70.23% to 71.43% on CiteSeer, while maintaining controlled ASR values. The cosine+dominant strategy combination shows improved robustness compared to individual cosine and dominant strategies, yet still exhibits moderate vulnerability with ASR values up to 30.0% on Cora and 10.0% on CiteSeer. The accuracy performance is intermediate, achieving 76.89% to 79.30% on Cora and 67.52% to 69.47% on CiteSeer, indicating that this combination may not fully address the inherent limitations of the cosine strategy.

Our proposed SPROUT-GNN framework demonstrates superior performance across both datasets. On Cora, it achieves consistently high accuracy (86.88% to 88.35%) while maintaining low ASR values (0.0% to 20.0%). On CiteSeer, SPROUT-GNN shows the best overall performance with accuracy ranging from 71.43% to 72.78% and ASR values from 0.0% to 6.67%.

The combination of different strategies generally outperforms individual approaches, demonstrating the complementary nature of different feature extraction policies. While individual strategies may offer better robustness in specific scenarios, combined approaches provide a superior balance between defensive capabilities and classification accuracy. The effectiveness of different strategies varies between datasets, with Cora generally showing higher accuracy but potentially higher vulnerability in certain configurations compared to CiteSeer. The superior performance of SPROUT-GNN confirms that our framework effectively leverages the strengths of different subgraph feature extraction policies while mitigating their individual weaknesses, resulting in optimal performance across different experimental conditions.

VIII. CONCLUSION

This paper presents SPROUT-GNN, a novel framework for enhancing the robustness of GNNs against adversarial attacks through strategic subgraph feature extraction. Our approach addresses a critical vulnerability in existing GNN architectures by integrating multiple feature extraction policies that collectively improve both defensive capabilities and classification accuracy. SPROUT-GNN provides built-in defense mechanisms based on two state-of-the-art existing defense methods. However, instead of applying them as a post-hoc defense, we integrate this capability directly into the model architecture, creating an inherently robust system. Leveraging HOGNN, SPROUT-GNN combines ego k-hop neighborhood extraction, outlier detection-based strategy selection, and cosine similarity-based feature aggregation. Through comprehensive experiments on standard benchmark datasets, we demonstrate that our framework consistently outperforms individual strategies and their pairwise combinations. The ablation studies reveal that while individual strategies may excel in specific scenarios, the integrated approach effectively mitigates their respective limitations while preserving their strengths. Future work includes extending the evaluation to larger datasets and investigating the theoretical foundations of multi-strategy integration to provide insights into robust GNN design principles.

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